

Literature Status: A Quantitative Local Advance on Planar $W^{2,1}$ Diffeomorphic Approximation

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Status: literature-implied answer to a partial subcase; the full problem remains open. Campbell–Hencl ask for an extension step that would turn their approximation outside a small exceptional set into a global $W^{2,1}$ diffeomorphic approximation. D’Onofrio, arXiv:2604.04592, proves a quantitative localized smoothing theorem for a substantially more structured piecewise-quadratic class, but explicitly leaves both a finite-singularity completion step and the general geometric approximation step unresolved.

1 The source program

D. Campbell and S. Hencl, *Approximation of planar Sobolev $W^{2,1}$ homeomorphisms by piecewise quadratic homeomorphisms and diffeomorphisms*, arXiv:2008.05736, prove that a planar $W^{2,1}$ homeomorphism with positive Jacobian almost everywhere can be approximated by a smooth diffeomorphism outside a locally finite union of squares of arbitrarily small area. On PDF page 4 they state their next step [1]:

Source program. *Obtain a second-order analogue of the key planar extension theorem used in the $W^{1,1}$ theory, thereby filling the exceptional squares and obtaining the full $W^{2,1}$ approximation theorem. A further goal is to remove the assumption $J_f > 0$ almost everywhere.*

The first clause is the target considered here.

2 The 2026 partial advance

L. D’Onofrio, *$W^{2,1}$ approximation of planar Sobolev homeomorphisms by smooth diffeomorphisms*, arXiv:2604.04592, begins by stating that no general $W^{2,1}$ approximation theorem is known. Its Assumption 5.1 (PDF page 14) concerns a bounded polygonal domain with a finite conforming rectangular partition and a map g which is

- quadratic on each cell and C^1 -compatible across every interface;
- a homeomorphism onto its image;
- quantitatively nondegenerate, with $\det Dg \geq \lambda > 0$; and
- quantitatively injective, with $|g(x) - g(y)| \geq m|x - y|$ for all x, y .

Theorem 5.5 (PDF pages 17–20) then gives the following [2].

Later partial theorem. *For every $\delta > 0$, there is a C^1 map $\tilde{g}_\delta$ with*

$$\|\tilde{g}_\delta - g\|_{W^{2,1}} < \delta, \quad \|D\tilde{g}_\delta - Dg\|_{L^\infty} < \delta, \quad \det D\tilde{g}_\delta \geq \lambda/2.$$

Moreover,

$$|\tilde{g}_\delta(x) - \tilde{g}_\delta(y)| \geq (m - Q_\Omega \delta)|x - y|,$$

so it is injective for sufficiently small δ , and it is smooth away from arbitrarily small neighborhoods of the finite set of endpoints of the interior edges.

This resolves a genuine local analytic part of the Campbell–Hencl program: interface and vertex regularization can preserve $W^{2,1}$ control, positive Jacobian, and global injectivity under explicit quantitative hypotheses.

3 Why this is not the full answer

There are two strict gaps between the later theorem and the source problem. First, the source’s approximation outside small squares does not furnish a global finite, C^1 -compatible, uniformly bi-Lipschitz piecewise-quadratic map satisfying Assumption 5.1. The 2026 paper itself calls construction of such structured approximants the remaining “global geometric approximation problem.”

Second, Theorem 5.5 is smooth only off small neighborhoods of finitely many edge endpoints. Its Theorem 6.2 obtains globally smooth injective maps only after imposing the separate, unproved Assumption 6.1, which is precisely a localized completion assertion. Thus arXiv:2604.04592 is a substantial partial advance and an explicit reduction, not a resolution of the general question. No new mathematical result is claimed in this packet.

References

- [1] D. Campbell and S. Hencl, *Approximation of planar Sobolev $W^{2,1}$ homeomorphisms by piecewise quadratic homeomorphisms and diffeomorphisms*, ESAIM Control Optim. Calc. Var. **27** (2021), article 26; arXiv:2008.05736.
- [2] L. D’Onofrio, *$W^{2,1}$ approximation of planar Sobolev homeomorphisms by smooth diffeomorphisms*, arXiv:2604.04592 (2026).